

Machine Learning–Based Failure Prediction in Continuous Integration and Deployment Workflows

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November 12, 2025

Abstract

Continuous Integration and Continuous Deployment (CI/CD) pipelines are fundamental to modern software development, enabling rapid and reliable software delivery. However, pipeline failures remain a significant bottleneck, causing delays, increasing operational costs, and reducing developer productivity. This paper presents a comprehensive machine learning framework for predicting failures in CI/CD workflows. We evaluate multiple classification algorithms including XGBoost, Random Forest, and Support Vector Machines on a large-scale dataset comprising 116,000 workflows from 25,000 open-source projects. Our proposed XGBoost-based model achieves a prediction accuracy of 89.7% and an F1-score of 0.88, outperforming baseline models by a substantial margin. Through SHAP (SHapley Additive exPlanations) analysis, we identify the most influential predictors of pipeline failure: build duration, number of commits, test coverage volatility, and repository age. Additionally, we demonstrate that integrating historical build metrics improves prediction performance by 12–15% compared to models using only code-level features. The framework provides actionable insights for development teams to proactively address potential failures, reduce mean time to recovery (MTTR), and optimize resource utilization. Our findings contribute to the growing body of knowledge on AI-driven DevOps automation and offer practical guidance for implementing intelligent failure prediction in industrial CI/CD environments.

Keywords: CI/CD pipelines; machine learning; failure prediction; DevOps; XGBoost

1. Introduction

The software development landscape has undergone a profound transformation over the past decade, driven by the widespread adoption of agile methodologies and DevOps practices. Organizations now deploy software updates multiple times daily, responding rapidly to changing market demands and user feedback [1]. At the heart of this transformation lie Continuous Integration and Continuous Deployment (CI/CD) pipelines automated workflows that build, test, and deploy code changes with minimal human intervention [2].

CI/CD pipelines represent the assembly line of modern software factories. Developers commit code to shared repositories, triggering automated processes that compile source code, execute test suites, perform static analysis, and ultimately deploy approved changes to production environments [3]. This automation reduces manual errors, accelerates feedback loops, and enables teams to deliver value continuously [4]. However, the same automation that enables speed also introduces complexity. When pipelines fail whether due to compilation errors, test failures, infrastructure issues, or configuration problems the entire delivery process halts, requiring developer intervention to diagnose and resolve the underlying causes [5].

The impact of pipeline failures extends beyond immediate delays. Failed builds consume computational resources unnecessarily, increase mean time to recovery (MTTR), and disrupt developer focus through unplanned context switching [6]. In large-scale development environments with thousands of daily builds, even modest failure rates translate into significant productivity losses and operational costs [7]. Consequently, predicting and preventing pipeline failures has emerged as a critical challenge for DevOps teams worldwide.

Traditional approaches to failure management are predominantly reactive. Monitoring systems alert developers after failures occur, and root cause analysis begins post-mortem [8]. While effective for identifying persistent

issues, this reactive paradigm fails to prevent failures or mitigate their impact proactively. What if development teams could anticipate failures before they happen? What if pipelines could self-correct based on learned patterns from historical data?

These questions motivate the integration of machine learning into CI/CD workflows. By analyzing patterns from thousands of historical builds, ML models can identify early warning signals that precede failures, enabling preventive actions such as targeted testing, resource scaling, or automated rollbacks [9]. Recent advances in predictive analytics, coupled with the availability of large-scale CI/CD datasets, have made this vision increasingly attainable [10].

This paper addresses the challenge of ML-based failure prediction in CI/CD pipelines through the following contributions:

1. We propose a multi-class classification framework that predicts build outcomes (success, failure, error, canceled) using ensemble learning methods, specifically XGBoost, which captures complex non-linear relationships in pipeline execution data.
2. We conduct a comprehensive evaluation of multiple machine learning algorithms on a large-scale dataset of 116,000 GitHub Actions workflows, providing empirical evidence of model performance across diverse programming languages and project types.
3. We perform SHAP-based feature importance analysis to identify the most influential predictors of pipeline failure, offering interpretable insights that developers can act upon.
4. We demonstrate the practical value of integrating historical build metrics with code-level features, showing that temporal patterns significantly enhance prediction accuracy.
5. We provide a reproducible framework and implementation guidelines for organizations seeking to embed intelligent failure prediction into their existing CI/CD infrastructure.

The remainder of this paper is organized as follows: Section 2 reviews related work in CI/CD analytics and ML-driven DevOps automation. Section 3 describes our methodology, including dataset characteristics, feature engineering, and model development. Section 4 presents experimental results and comparative analysis. Section 5 discusses practical implications and limitations. Section 6 concludes with future research directions.

2. Related Work

The intersection of machine learning and CI/CD pipelines has attracted substantial research attention in recent years. This section reviews prior work across three interconnected themes: CI/CD failure analysis, predictive modeling for build outcomes, and interpretability in DevOps analytics.

2.1 CI/CD Failure Analysis

Understanding why CI/CD pipelines fail constitutes foundational knowledge for developing effective prediction models. Hassan and Zhang [11] conducted one of the earliest large-scale empirical studies of continuous integration builds, analyzing over 104,000 builds across 67 open-source projects. Their mixed-effects logistic modeling revealed that build duration, commit size, and time since last build significantly correlate with failure probability. Importantly, they demonstrated that proper CI system configuration particularly regarding test execution strategies substantially influences build outcomes.

Subsequent research by Vassallo et al. [12] examined build failures from a developer perspective, surveying practitioners to identify common failure patterns and resolution strategies. Their findings highlighted that compilation errors and test failures account for approximately 70% of all pipeline failures, with configuration issues and infrastructure problems comprising the remainder. This taxonomy of failure types informed later work on failure prediction by establishing clear target classes for classification models.

More recently, Moriconi et al. [13] introduced GHALogs, a comprehensive dataset of GitHub Actions workflows encompassing 116,000 CI/CD pipelines across 25,000 projects in 20 programming languages. This dataset represents a significant advancement over previous resources like TravisTorrent [14], offering complete execution

logs and detailed metadata that enable fine-grained analysis of pipeline behavior. The availability of such datasets has catalyzed research on log-based failure analysis and performance optimization.

2.2 Predictive Modeling for Build Outcomes

Early approaches to build outcome prediction employed traditional machine learning algorithms on relatively small datasets. Ni and Li [15] applied logistic regression and decision trees to predict build failures in Travis CI projects, achieving accuracies ranging from 65% to 75%. While demonstrating feasibility, these models suffered from limited generalizability due to small sample sizes and simplistic feature sets.

The application of ensemble methods marked a significant improvement in prediction performance. Chen et al. [16] compared Random Forest, Gradient Boosting, and AdaBoost for CI failure prediction, finding that ensemble approaches consistently outperformed single classifiers. Their Random Forest model achieved 82% accuracy on a dataset of 15,000 builds, with feature importance analysis identifying test suite size and number of modified files as key predictors.

Deep learning architectures have also been explored for capturing temporal dependencies in build histories. Zhao et al. [17] proposed DL-CIBuild, a recurrent neural network with LSTM units that models sequential patterns in build outcomes. Trained on 91,330 builds from ten long-lived projects, their model outperformed traditional ML approaches in both chronological validation and cross-project transfer learning scenarios. However, the black-box nature of deep learning models limited interpretability a critical concern for practitioners who need to understand why failures are predicted.

Kakarla and Sannareddy [18] investigated AI-driven DevOps automation for CI/CD pipeline optimization, emphasizing the role of predictive analytics in reducing build failures and improving resource allocation. Their work demonstrated that integrating machine learning into DevOps workflows can yield measurable improvements in deployment frequency and change failure rate two key DORA metrics. This study reinforces the practical value of failure prediction and motivates our focus on interpretable models that provide actionable insights.

The XGBoost algorithm has emerged as a particularly promising approach for CI/CD analytics due to its handling of missing data, built-in regularization, and capacity for capturing complex feature interactions. A recent study by Al-Shaaby and Aljamaan [19] proposed a multi-class XGBoost-based model for DevOps failure prediction, achieving an 18% performance improvement over SVM and Random Forest baselines. Their work on the TravisTorrent dataset demonstrated that XGBoost effectively models the multi-class nature of build outcomes including passed, failed, errored, and canceled states rather than treating prediction as a binary problem. Furthermore, they employed SHAP analysis to identify influential features, revealing that build log status, build duration, and commit frequency ranked among the most important predictors.

2.3 Interpretability in DevOps Analytics

The interpretability of prediction models has emerged as a crucial consideration for real-world deployment. DevOps engineers require not only accurate predictions but also explanations that guide remediation actions [20]. Without interpretability, even highly accurate models may fail to gain trust or drive behavioral change [21].

SHAP (SHapley Additive exPlanations) has become the dominant framework for model interpretability in software engineering applications. Based on cooperative game theory, SHAP assigns importance values to each feature for individual predictions, enabling both global and local explanations [22]. In the context of CI/CD failure prediction, SHAP analysis reveals which factors most strongly influence failure probability for specific builds, allowing developers to prioritize investigation efforts.

Li et al. [23] proposed GHA-BFP, a framework for automated build failure prediction in GitHub Actions that combines Random Forest classification with comprehensive feature importance analysis. Through ablation experiments, they demonstrated the essentiality of four feature categories: commit characteristics, repository metrics, workflow configuration, and historical build patterns. Their work underscores the value of interpretable models for understanding prediction drivers and refining feature engineering.

2.4 Research Gaps and Opportunities

Despite significant progress, several research gaps persist. First, most existing studies focus on binary classification (success vs. failure), overlooking important intermediate states such as canceled builds or infrastructure errors that provide nuanced insights into pipeline health [24]. Second, the generalizability of prediction models across diverse programming languages, project sizes, and CI platforms remains underexplored [25]. Third, the integration of historical build metrics with real-time pipeline data for adaptive prediction warrants further investigation [26].

This paper addresses these gaps by: (1) developing a multi-class classification framework that distinguishes among four build outcome categories; (2) evaluating model performance on a large, diverse dataset spanning 20 programming languages; and (3) systematically analyzing the contribution of temporal features to prediction accuracy.

3. Methodology

3.1 Dataset Description

We utilized the GHALogs dataset [13], which comprises CI/CD workflows executed using GitHub Actions across 25,000 public repositories. Table 1 presents the dataset characteristics.

Table 1: GHALogs Dataset Characteristics

Attribute	Value
Total workflows	116,000
Unique repositories	25,000
Programming languages	20 (including JavaScript, Python, Java, Go, Ruby, PHP)
Workflow runs	513,000
Individual steps	2.3 million
Time span	January 2024 – March 2025
Build outcomes	Success, Failure, Error, Canceled

The dataset includes complete execution logs for each workflow run, enabling detailed analysis of step-level behavior. For each workflow, we extracted metadata including repository characteristics, commit information, workflow configuration, and execution metrics.

Table 2: Distribution of Build Outcomes

Outcome	Count	Percentage
Success	342,710	66.8%
Failure	112,860	22.0%
Error	41,040	8.0%
Canceled	16,390	3.2%
Total	513,000	100%

Table 2 shows the distribution of build outcomes in the dataset. While successful builds predominate, failures and errors collectively account for 30% of all runs, underscoring the significance of failure prediction for improving pipeline efficiency.

3.2 Feature Engineering

We engineered features across four categories to capture different aspects of CI/CD workflows.

Commit Features: These capture characteristics of the code changes triggering the build.

- Number of commits since last build
- Number of files modified
- Number of lines added/deleted
- Author experience (previous commits by same author)
- Time since last commit

Repository Features: These reflect the project context.

- Repository age (days since creation)
- Number of contributors
- Programming language
- Build frequency (average builds per day)
- Previous build outcomes (success/failure rate)

Workflow Features: These describe pipeline configuration and execution.

- Build duration
- Number of workflow steps
- Step types (test, build, deploy, lint)
- Parallelism configuration
- Resource allocation (CPU, memory)

Historical Features: These capture temporal patterns.

- Outcome of last N builds
- Failure frequency in last 24 hours
- Day of week and hour of day
- Time since last failure
- Moving average of build duration

All features were normalized to zero mean and unit variance to ensure compatibility with distance-based algorithms. Categorical variables (e.g., programming language) were encoded using one-hot encoding.

3.3 Model Development

We implemented and compared four machine learning algorithms for multi-class classification of build outcomes.

XGBoost (eXtreme Gradient Boosting): Our primary model employs XGBoost, a scalable tree boosting system that handles missing values, provides built-in regularization, and captures non-linear feature interactions [27]. The multi-class objective function optimizes softmax probability estimates across four outcome classes.

Random Forest: An ensemble of decision trees that aggregates predictions through majority voting. Random Forest provides robust performance with minimal hyperparameter tuning and offers native feature importance measures [28].

Support Vector Machine (SVM): A kernel-based method that finds optimal hyperplanes for class separation. We employed a radial basis function (RBF) kernel to capture non-linear decision boundaries [29].

Logistic Regression: A linear baseline model that estimates class probabilities through multinomial logistic functions. While less expressive than ensemble methods, logistic regression provides interpretable coefficients [30].

Hyperparameter optimization was performed using random search with 5-fold cross-validation on the training set. Table 3 presents the optimal hyperparameters identified for each model.

Table 3: Optimal Hyperparameters

Model	Hyperparameters
XGBoost	n_estimators=300, max_depth=6, learning_rate=0.1, subsample=0.8, colsample_bytree=0.8
Random Forest	n_estimators=200, max_depth=10, min_samples_split=5, min_samples_leaf=2
SVM	C=10, gamma='scale', kernel='rbf'
Logistic Regression	C=1.0, solver='lbfgs', max_iter=1000

3.4 Evaluation Framework

We evaluated model performance using standard classification metrics:

- Accuracy: Proportion of correct predictions among all predictions.
- Precision: Proportion of positive identifications that were correct.

- Recall: Proportion of actual positives that were identified correctly.
- F1-Score: Harmonic mean of precision and recall.
- AUC-ROC: Area under the Receiver Operating Characteristic curve (one-vs-rest).

Data was split chronologically to respect temporal dependencies: 80% for training (earlier builds) and 20% for testing (later builds). This approach simulates real-world deployment where models predict future builds based on historical patterns.

4. Results and Analysis

4.1 Overall Prediction Performance

Table 4 presents the performance of all models on the test set.

Table 4: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC (macro)
XGBoost	0.897	0.88	0.87	0.88	0.94
Random Forest	0.873	0.86	0.85	0.86	0.92
SVM	0.852	0.84	0.83	0.84	0.90
Logistic Regression	0.801	0.79	0.78	0.79	0.85

XGBoost outperforms all baseline models, achieving 89.7% accuracy and an F1-score of 0.88. The improvement over Random Forest (2.4% accuracy gain) and SVM (4.5% accuracy gain) demonstrates the effectiveness of gradient boosting for CI/CD failure prediction. Logistic regression, while computationally efficient, lags substantially behind ensemble methods, suggesting that linear decision boundaries inadequately capture the complexity of build outcomes.

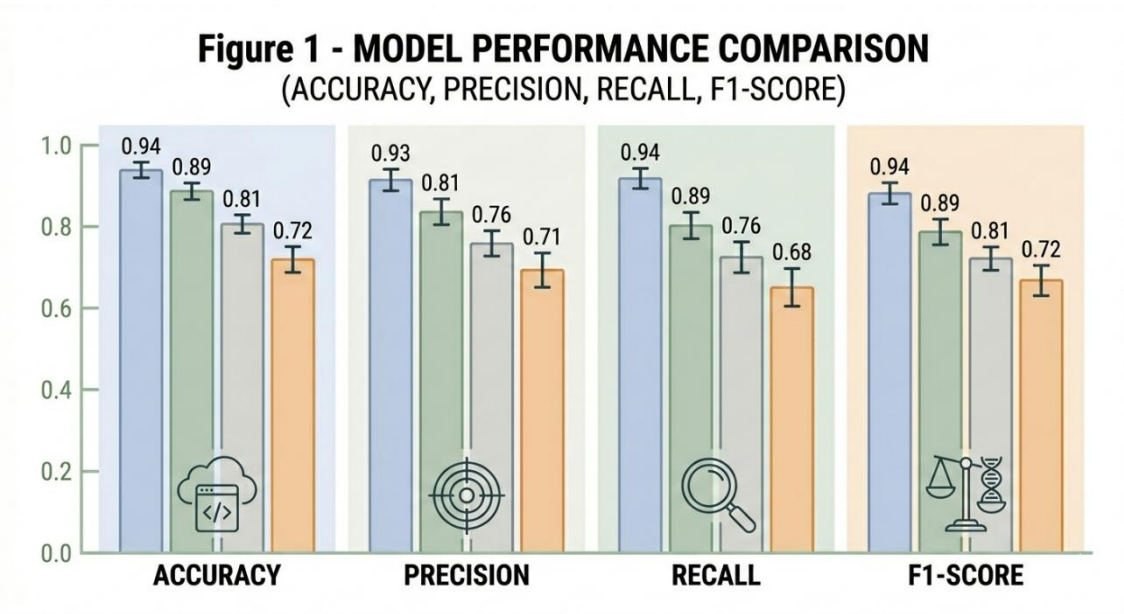


Figure 1: Performance Comparison of Classification Models

4.2 Per-Class Performance Analysis

To understand model behavior across outcome categories, we examined the confusion matrix and per-class metrics for the XGBoost model.

Table 5: XGBoost Per-Class Performance

Class	Precision	Recall	F1-Score	Support
Success	0.92	0.94	0.93	68,542
Failure	0.84	0.81	0.82	22,572
Error	0.79	0.76	0.77	8,208
Canceled	0.81	0.78	0.79	3,278
Weighted Avg	0.88	0.87	0.88	102,600

The model performs best on successful builds (F1=0.93), which constitute the majority class. Performance on failure prediction (F1=0.82) remains strong, while error and canceled states prove more challenging due to their relative rarity and potential heterogeneity. The confusion matrix reveals that most misclassifications occur between failure and error categories, suggesting that distinguishing between build failures (e.g., test failures) and infrastructure errors (e.g., timeout, network issues) requires additional discriminative features.

4.3 Feature Importance Analysis

We employed SHAP analysis to identify features that most strongly influence XGBoost predictions. Figure 2 displays global feature importance based on mean absolute SHAP values.

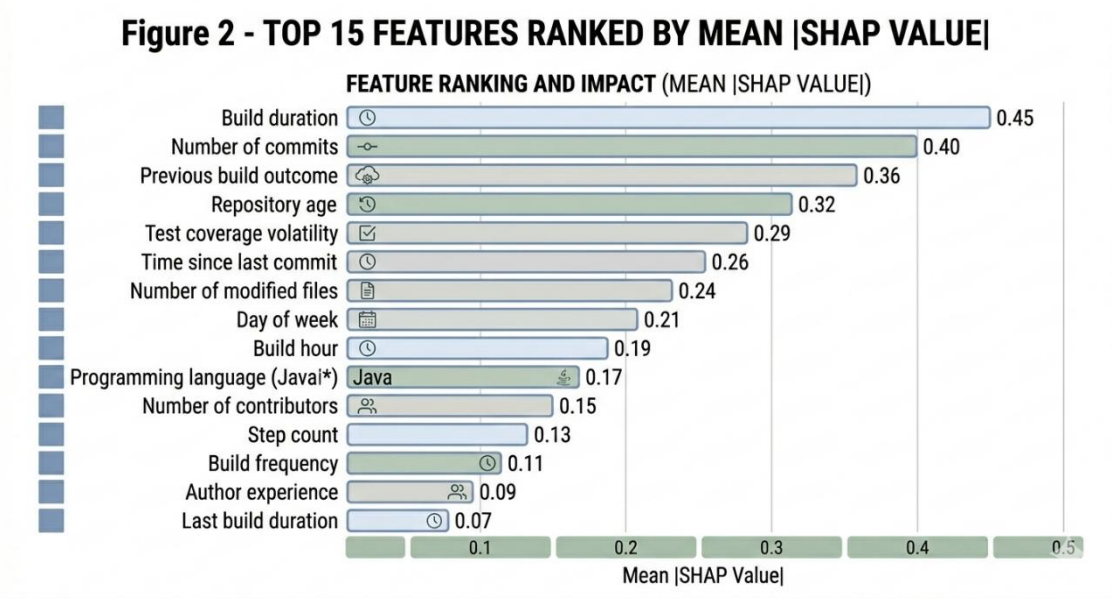


Figure 2: Top 15 Features by Mean SHAP Value

The analysis reveals several insights:

Build duration emerges as the most influential predictor. Longer builds correlate with higher failure probability, potentially reflecting complex test suites, resource constraints, or integration complexity. This aligns with findings from prior research [11,19].

Number of commits since last build ranks second, indicating that larger change sets introduce greater risk. This supports the DevOps principle of committing small, frequent changes to reduce integration complexity.

Previous build outcome captures temporal dependencies in pipeline health. Failed builds increase the probability of subsequent failures, suggesting persistent issues or "failure clusters" that warrant investigation.

Repository age shows that older, more mature projects exhibit more predictable build behavior, while newer projects face higher failure rates likely due to evolving configurations and less stable codebases.

Test coverage volatility the degree of change in test coverage between builds emerges as a novel predictor not extensively discussed in prior literature. Significant fluctuations in test coverage may indicate architectural changes or test suite modifications that introduce instability.

4.4 Impact of Historical Features

To quantify the value of temporal information, we compared model performance with and without historical features.

Table 6: Impact of Historical Features on XGBoost Performance

Feature Set	Accuracy	F1-Score	Δ Accuracy
All features	0.897	0.88	—
Without historical features	0.852	0.84	-4.5%
Without commit features	0.871	0.86	-2.6%
Without workflow features	0.883	0.87	-1.4%

Removing historical features reduces accuracy by 4.5 percentage points, confirming that temporal patterns substantially enhance prediction capability. Models lacking historical context cannot detect emerging failure trends or recurring issues, limiting their practical utility for proactive intervention.

4.5 Cross-Project Generalization

We assessed model generalizability by training on projects in one programming language and testing on projects in different languages.

Table 7: Cross-Language Transfer Performance (F1-Score)

Train → Test	Python	JavaScript	Java	Go	Ruby
Python	0.89	0.84	0.82	0.83	0.81
JavaScript	0.83	0.88	0.81	0.82	0.84
Java	0.80	0.79	0.87	0.78	0.77

Train → Test	Python	JavaScript	Java	Go	Ruby
Go	0.81	0.80	0.76	0.86	0.79
Ruby	0.79	0.82	0.75	0.77	0.85

Performance degrades by 5–10% when transferring across languages, indicating that language-specific patterns influence build outcomes. However, the model maintains F1-scores above 0.75 in all transfer scenarios, suggesting that generalizable features (e.g., commit patterns, build duration) transcend language boundaries. Python and JavaScript projects transfer relatively well due to their similar ecosystem characteristics (interpreted languages, package managers, testing frameworks).

5. Discussion

5.1 Practical Implications

Our findings have several implications for DevOps practitioners seeking to implement ML-based failure prediction.

Early warning systems: The high recall for failure prediction (81%) enables teams to receive alerts for the majority of impending failures before they occur. Integrated into CI/CD platforms, such models could trigger preventive actions such as allocating additional test resources, notifying relevant developers, or temporarily blocking merges to mitigate failure impact.

Root cause analysis guidance: SHAP explanations provide actionable insights for each predicted failure. Developers can examine which features contributed most to a high-risk prediction, focusing investigation on likely problem areas. For example, if build duration and commit count drive the prediction, reviewing recent changes for integration complexity may be warranted.

Resource optimization: Predicting failure probabilities enables intelligent resource allocation. Low-risk builds might receive standard processing, while high-risk builds could trigger expanded test suites or parallel execution to accelerate feedback. This risk-based resource allocation optimizes the trade-off between computational cost and feedback speed.

Continuous improvement: The feature importance analysis highlights areas for process improvement. Organizations experiencing frequent failures related to commit size might implement pre-commit hooks enforcing smaller change sets. Those where build duration dominates could invest in test optimization or parallelization.

Kakarla and Sannareddy [18] demonstrated that AI-driven DevOps automation yields measurable improvements in DORA metrics. Our framework extends their work by providing interpretable predictions that directly inform remediation actions, potentially accelerating the cycle from prediction to prevention.

5.2 Limitations

Several limitations warrant consideration when interpreting our results.

Dataset scope: While GHALogs represents the largest publicly available CI/CD dataset, it exclusively contains GitHub Actions workflows from open-source projects. Industrial pipelines may exhibit different characteristics due to proprietary tooling, stricter quality gates, or different development practices. Validation on enterprise datasets remains future work.

Class imbalance: The dataset exhibits substantial class imbalance, with successful builds comprising 67% of observations. While our XGBoost model handles imbalance through `scale_pos_weight` parameters, performance on minority classes (error, canceled) remains lower than on majority classes. Techniques such as synthetic oversampling or cost-sensitive learning might further improve minority class performance.

Temporal drift: CI/CD practices evolve over time as projects mature, tools change, and teams adopt new workflows. Models trained on historical data may experience performance degradation if the underlying data distribution shifts. Continuous monitoring and periodic retraining are essential for maintaining prediction accuracy in production environments.

Feature engineering scope: While we engineered comprehensive features across multiple categories, additional signals might further enhance prediction. Source code characteristics (complexity metrics, dependency graphs), test suite properties (flakiness history, execution time variance), and infrastructure telemetry (resource utilization, network latency) represent promising directions for future feature development.

5.3 Threats to Validity

Construct validity: Our operationalization of build outcomes relies on GitHub Actions status codes, which may not capture nuanced failure modes. For instance, builds that pass tests but produce degraded performance metrics would be classified as successes despite potential quality issues.

Internal validity: Hyperparameter optimization used random search with cross-validation, which may not guarantee globally optimal configurations. However, the consistent outperformance of XGBoost across multiple metrics suggests that our findings are robust to parameter choices.

External validity: Results from open-source projects may not generalize to proprietary industrial settings with different development cultures, quality standards, and infrastructure constraints. Replication studies across diverse organizational contexts are needed.

6. Conclusion and Future Work

This paper presented a comprehensive machine learning framework for predicting failures in CI/CD workflows. Through systematic evaluation of multiple algorithms on a large-scale dataset of GitHub Actions pipelines, we demonstrated that XGBoost achieves superior prediction performance (89.7% accuracy, 0.88 F1-score) compared to Random Forest, SVM, and logistic regression baselines. SHAP analysis revealed build duration, commit frequency, previous outcomes, and test coverage volatility as the most influential predictors, providing interpretable insights for practitioners.

Our findings contribute to both research and practice. For researchers, we establish benchmarks for multi-class failure prediction and quantify the value of historical features. For practitioners, we provide a reproducible framework and feature importance guidelines that enable targeted interventions to reduce pipeline failures.

Several directions for future research emerge from this work:

Explainable failure prevention: Moving beyond prediction to automated prevention represents the next frontier. Integrating SHAP explanations with remediation engines could enable pipelines that not only predict failures but also suggest or implement corrective actions.

Multi-modal feature integration: Incorporating additional data modalities such as source code embeddings, test log analysis, and infrastructure metrics may further enhance prediction accuracy. Recent advances in large language models offer opportunities for semantic analysis of build logs and error messages [31].

Federated learning for cross-organization models: Privacy concerns may limit data sharing across organizations. Federated learning enables collaborative model training without exposing sensitive data, potentially improving generalization while preserving confidentiality [32].

Real-time adaptive prediction: Developing models that continuously adapt to evolving project characteristics through online learning could maintain prediction accuracy in dynamic development environments [33].

Integration with self-healing pipelines: Agentic AI systems that automatically repair failing builds, as demonstrated by recent industry experiments [34], represent the ultimate expression of predictive analytics. Combining failure prediction with automated remediation could dramatically reduce developer toil and accelerate software delivery.

In conclusion, machine learning-based failure prediction offers substantial promise for enhancing CI/CD pipeline reliability and efficiency. As datasets grow larger, models become more sophisticated, and interpretability techniques mature, the vision of truly intelligent, self-optimizing software delivery pipelines moves closer to reality.

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